https://doi.org/10.1038/s41567-026-03411-3

A digital twin should make a risky claim

Manlio De Domenico, Valeria d'Andrea, Marco Grassia, Giuseppe Mangioni & Barbara Di Camillo

Check for updates

As the term 'digital twin' spreads across science and technology, it risks losing scientific meaning. Such a twin should not denote just a digital replica, but a computational model that can be updated, tested against intervention and shown to fail.

The term digital twin is becoming too cheap. It is routinely used to indicate dashboards, monitoring platforms, simulators, predictive models and artificial intelligence systems attached to data streams. These are not the same objects, and the shared label blurs the difference between a weak claim and a strong one. For physicists, that distinction matters. A digital twin should not simply mean that a system has been digitized, visualized or forecast. It should mean that a model is tightly linked to a specific real-world system and evolves alongside it[1]. What ultimately defines a digital twin is that the model can inform decisions and guide actions in a changing environment (Fig. 1).

That is a risky claim, and it is why the term is worth defending. A digital twin should track a specific system over time. New observations should update the model's estimate of the system's state, allowing possible interventions to be compared. Uncertainty should, by design, be part of the output rather than a footnote. A digital twin's credibility can be lost as the system, or the conditions under which it is observed, changes. Accordingly, a digital twin is not merely a virtual replica; it constitutes a computational framework that explicitly makes model errors observable, quantifiable and open to external scrutiny.

This is also why digital twins should interest physicists. The hard part is the science of reduction under constraint: deciding what can be left out without losing what matters for action, estimating hidden variables from partial observations and keeping a model useful after an intervention has changed the system it was built to follow.

An emblematic example comes from cardiovascular medicine. The most convincing progress there has come from closing a loop: patient-specific measurements constrain a model, the model forecasts clinically relevant trajectories, alternative actions can be compared and subsequent data update the estimate of the patient's state[2]. The lesson extends beyond this specific application: a digital twin becomes scientifically interesting when it stops being a static representation and starts making revisable, decision-relevant claims.

A cell provides a useful test case for what turns a model into a twin. A mechanistic or statistical model of cellular dynamics is not, in itself, a digital twin. The model becomes a digital twin only when it is linked to a specific evolving cellular system and updated as new measurements arrive. This criterion already excludes much of what is often casually presented as twinning, from individual cells to entire ecosystems. A generic simulator, a one-shot predictor or a data platform that gathers streams without instantiating an updating model is not sufficient.

However, even this minimal condition is demanding. In a cell, for example, the target of twinning over time is not obvious and is hard to delimit. Measurements are sparse, noisy, delayed, invasive or even destructive, with relevant variables spanning molecules,

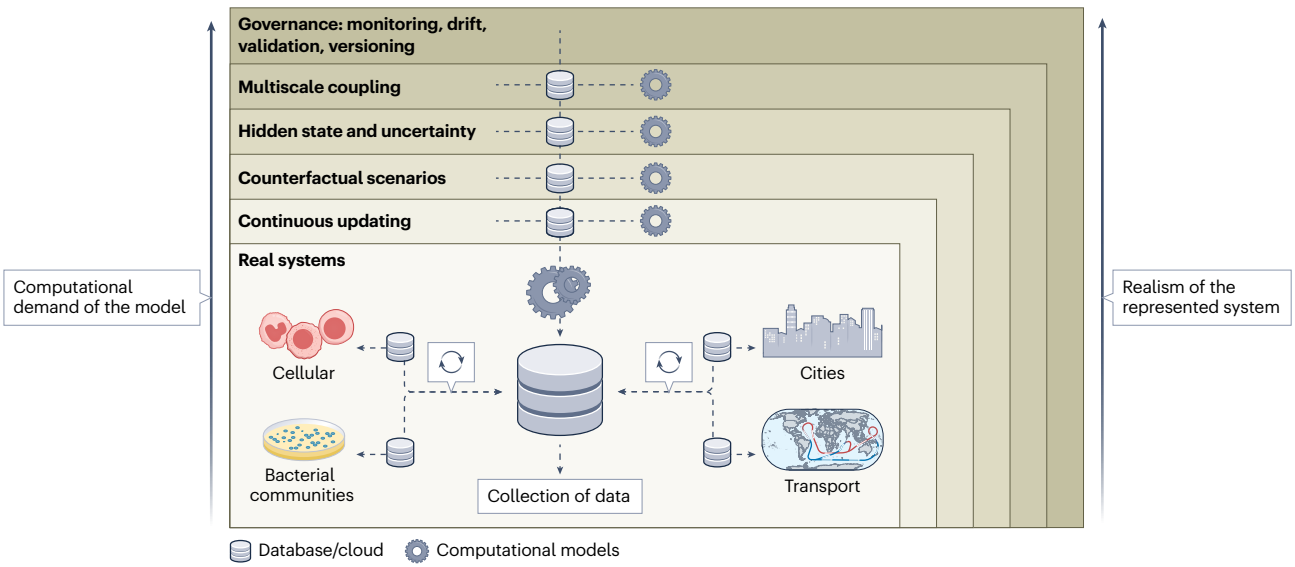


Fig. 1 | Digital twins as progressively stronger claims. A digital twin is instantiated from a well defined physical or operational system that is instrumented with measurements and associated with an adaptively updated model. Each successive layer in its architecture contributes an additional functional capability.

pathways and phenotypes. The system is complex, adaptive and far from equilibrium[3]. To call an updating model of cellular dynamics a digital twin is therefore to assert that the chosen observables can keep track of a changing counterpart rather than merely describe a class of possible cells.

Synchronization, the process of aligning a model with a specific real system using incoming measurements, is only the initial phase of the process. The promise of digital twins is to support choices among possible futures. In the case of a cell, this means comparing perturbations, for example, between different drug pulses, gene modifications or stimulation protocols. Once a model is used to compare interventions under explicit assumptions, the term digital twin begins to acquire explanatory importance.

At a much larger spatial scale, complex adaptive systems such as cities make the next difficulty unavoidable. No city is fully observable. Mobility traces, environmental sensors, power demand, hospital admissions, economic activity and administrative records are just fragments of a much larger dynamical state. Many variables that matter for intervention are often hidden, including behavioural response, vulnerability, compliance, bottlenecks, threshold effects and informal interactions. A municipal-scale platform capable of aggregating data streams in real time may enhance situational awareness, but real-time data accumulation does not, in itself, render the platform a digital twin. To function as a twin, a city model must estimate the changing state of the urban system and make clear which variables are observed, reconstructed and assumed[4].

The urban context also reveals another common ambiguity in the contemporary discourse on digital twins: complexity is often mistaken for realism. The usual image is that of a giant digital replica into which every subsystem is poured. However, increasing the level of detail does not inherently ensure the construction of a superior digital twin. The relevant scientific question is to identify which elements are essential to include for the specific decision problem under consideration[5]. Traffic management, heat exposure, epidemic response, infrastructure resilience and social inequality do not all require the same representation. Therefore, a good digital twin is often a selective model. Its methodological core lies in determining which degrees of freedom can be coarse-grained, under what conditions historical dependence must be preserved and which couplings cannot be neglected because they mediate the propagation of intervention effects across hierarchical scales. The goal is to decide what must remain visible for a decision to be trusted.

These considerations represent classical physical inquiries expressed in an operational framework: which variables are causally relevant for system behaviour, which can be justifiably neglected and which may re-emerge as stochastic fluctuations, long-term memory effects or unanticipated phenomena. Digital twins make these modelling choices explicit and integral to the decision-making process.

And yet the real challenge is only just beginning. Interventions change the system they are meant to control. For example, a traffic forecast may cause drivers to reroute, and populations respond strategically to policy recommendations or infrastructure degradation. It is well known that forecasts affect behaviour and that behaviour can invalidate forecasts[6]. Therefore, the reliability of a digital twin cannot be established once and assumed to persist in a context-independent manner. In adaptive systems, validation has a half-life[7], and the reliability of a digital twin must be maintained over time. High-stakes modelling communities have long addressed this problem through verification, validation and uncertainty-quantification frameworks that tie credibility to the context of use and to the consequences of error[8]. For digital twins of complex adaptive systems, reliability requires active maintenance. This includes monitoring for model and data drift[9], formal specification of update criteria, maintenance of comprehensive audit trails and version histories and systematic re-examination of underlying assumptions following interventions.

Recent advances in machine learning provide scalable tools to support these requirements[10]. In particular, physics-informed surrogate models can accelerate feature selection and enable rapid model recalibration and counterfactual evaluation, whereas automatic drift detection can identify distributional shifts in incoming observations. Explainable artificial intelligence techniques — whose outputs humans can understand and inspect — could further strengthen this pipeline by making the assumptions embedded in surrogate models transparent and auditable.

In conclusion, a digital twin should be characterized by the strength of the claims it makes, rather than by the sophistication of its digital representation or the size of its data infrastructure. Operational dashboards, simulation environments, data infrastructures and artificial intelligence systems can all be useful. However, they do not qualify as digital twins solely because they possess sensors or real-time software architectures.

The term digital twin should thus be reserved for models that are continuously updated in the light of new evidence, used to evaluate candidate interventions, subjected to potential empirical falsification and revised as boundary conditions evolve. If this methodological standard is relaxed or rendered ambiguous, the term may persist in usage, but its scientific content will be effectively nullified.

Manlio De Domenico [1,2,3,4] ✉, **Valeria d'Andrea** [1,4], **Marco Grassia** [5], **Giuseppe Mangioni** [5] **& Barbara Di Camillo** [6]
[1]Department of Physics and Astronomy "Galileo Galilei", University of Padua, Padua, Italy. [2]Padua Center for Network Medicine, University of Padua, Padua, Italy. [3]Padova Neuroscience Center, University of Padua, Padua, Italy. [4]Istituto Nazionale di Fisica Nucleare, Padua, Italy. [5]Department of Electric, Electronic and Computer Engineering, University of Catania, Catania, Italy. [6]Department of Information Engineering, University of Padua, Padua, Italy.
✉e-mail: manlio.dedomenico@unipd.it

Published online: 11 September 2026

References

1. Wright, L. & Davidson, S. *Adv. Model. Sim. Eng. Sci.* **7**, 13 (2020).
2. Waight, M. C. et al. *Circulation* **151**, 521–533 (2025).
3. De Domenico, M. et al. *npj Digit. Med.* **8**, 37 (2025).
4. Carrassi, A., Bocquet, M., Bertino, L. & Evensen, G. *Wiley Interdisc. Rev. Clim. Change* **9**, e535 (2018).
5. Levins, R. *Am. Sci.* **54**, 421–431 (1966).
6. Lucas, R. E. Jr Econometric policy evaluation: a critique. In *Carnegie-Rochester Conference Series on Public Policy* (ed. McCallum, B. T.) Vol. 1, 19–46 (North-Holland, 1976).
7. Oreskes, N., Shrader-Frechette, K. & Belitz, K. *Science* **263**, 641–646 (1994).
8. Oberkampf, W. L. & Trucano, T. G. *Prog. Aerospace Sci.* **38**, 209–272 (2002).
9. Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M. & Bouchachia, A. *ACM Comput. Surv.* **46**, 44 (2014).
10. Karniadakis, G. E. et al. *Nat. Rev. Phys.* **3**, 422–440 (2021).

Acknowledgements

M.D.D. thanks the Italian Ministry of University and Research (grant FIS00000158) for supporting this research.

Competing interests

The authors declare no competing interests.